



BELVENAR ANALYTICS

IBM BOB HACKATHON · MAY 2026 · LABLAB.AI

ROSETTA

Database onboarding, automated.

Connect a database. Get an interactive onboarding guide — instantly.

BUILT WITH



IBM BOB



CLAUDE

github.com/Raven-V1/rosetta · MIT License · Carlos V

Documentation rot costs real time.

Every enterprise database has the same problem: no documentation that a new developer can actually use. They spend their first two weeks asking everyone questions instead of building.

2 Weeks

lost per new hire
learning the schema

\$0

documentation
provided on day one

7 People

interrupted daily
to answer questions

One connection. One complete guide.



Schema Map

Interactive visual diagram of every table and relationship, grouped by domain function.



Glossary

Every table described in plain language. Searchable. Click any table for column-level detail.



First Queries

10–15 annotated SQL queries, runnable in-app against your live database with results inline.



Download

Full onboarding guide exported as a single Markdown file. Version-control it alongside your code.





Bob didn't assist. Bob built.

ARCHITECT MODE

Generated system architecture, module breakdown, data-flow diagram, and Pydantic schema definition.

CODE MODE

Wrote every page of the Streamlit app — connection logic, schema introspection, LLM prompt wrappers, PDF export.

REVIEW MODE

Ran agentic review on every module. All flagged issues addressed inline. Session log exported for judging.

CUSTOM MODE

Built a Bob custom mode for database analysis — flags missing descriptions, type inconsistencies, naming violations.

40

BOBCOINS CONSUMED
of 40 total budget



Seven pages. One database connection. No SQL expertise required.

FRAMEWORK

Python 3.11 + Streamlit

DATABASE

pyodbc / SQL Server 2025

AI ENGINE

IBM Bob + Groq LLaMA-3.1

SCHEMA VIZ

streamlit-agraph

EXPORT

Markdown + reportlab PDF

DEPLOY

Streamlit Community Cloud

01

Home / Connect

02

Spotlight

03

Overview

04

Schema Map

05

First Queries

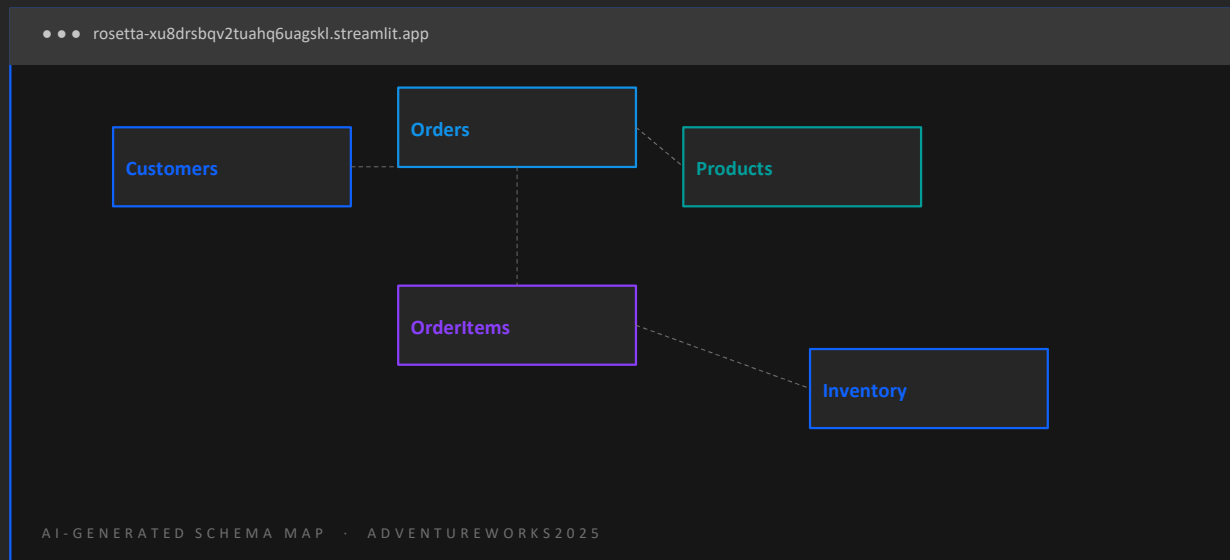
06

Glossary

07

Download

From connection string to onboarding guide.



1 min

from cold start
to full guide

7 pages

generated
automatically

0 SQL

knowledge
required



The Rosetta Stone translated a dead language. We translate your database.

Enterprise Onboarding

New analysts and developers become productive on day one instead of week three. SAP, Oracle, SQL Server — any relational database.

IBM Ecosystem Fit

IBM Bob drives the core AI. Groq LLaMA-3.1 handles natural language generation. Carbon Design tokens unify the UI. Native IBM stack, end to end.

Knowledge Preservation

Critical schema knowledge no longer lives only in the heads of senior developers. Export the guide, version-control it, keep it alive.

Open Source, MIT

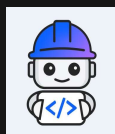
Every line of code is public at github.com/Raven-V1/rosetta. Fork it, contribute to it, deploy it inside your enterprise firewall.



ROSETTA

by Belvenar Analytics · IBM Bob Hackathon 2026

github.com/Raven-V1/rosetta



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CLAUDE

Carlos V